Retail Sales Analysis Report

Adithya C. V.

Dataset

[Retail Sales Dataset]

Objective

To perform real-world data analysis by cleaning and transforming raw data using Excel and creating interactive visualizations and dashboards in Power BI to generate meaningful insights and support data-driven decision-making.

Dataset Overview

The dataset contains retail business information related to orders, customers, products, sales, profit, shipping, payment methods, and customer satisfaction. It is a structured dataset used for performing data cleaning and transformation in Excel and data visualization and reporting in Power BI. The dataset helps analyze sales performance, customer behavior, profitability, order status, and delivery efficiency, enabling the creation of interactive dashboards and meaningful business insights for data-driven decision-making.

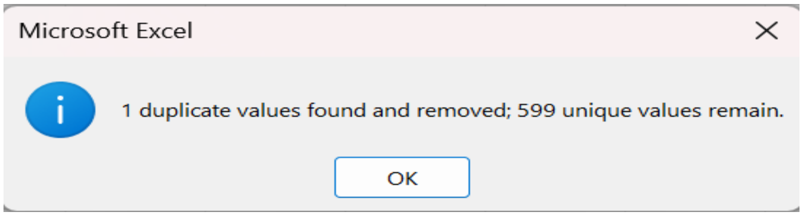
Table: Retail Sales: Contains 21 columns and 601 rows.

order_id	order_date	customer_	customer_	age	gender	region	city	product_c	product_n	quantity	unit_price	discount_	sales_amo	profit	shipping_	payment_	customer_	return_fla	order_stat	days_to_ship
ORD-0390	2024-08-15	CUST3260	Aarav Meh	39	Other	Central	Raipur	Groceries	Sugar	8	346.32		4043.14	289.18	56.04	Debit Card		TRUE	Returned	7
ORD-0219	2022-08-16	CUST0788	Sanjay Ver	28	Male	West	Mumbai	Beauty	Sunscreen	8	858.66	0.08	9147.31	5355.56	64.49	Debit Card	5	FALSE	Delivered	2
ORD-0134	2021-08-29	CUST4929	Pooja Shar	41	Other	West	Surat	Furniture	Sofa	4	25941.3	0.14	110157	36956.5	67.65	Debit Card	4	FALSE	Delivered	5
ORD-0207	2022-07-05	CUST0558	Priya Redc	29	Other	West	Ahmedaba	Furniture	Bed Frame	9	22071.9	0.17	262066	66645	54.47	EMI	3	FALSE	Delivered	4
ORD-0028	2020-05-07	CUST5803	Ritu Sharm	9	Male	Central	Nagpur	Groceries	Cooking O	1	320.06	0.31	323.95	31.38	48.81	Net Bankir	4	FALSE	Pending	4
ORD-0395	02/09/2024	CUST1718	Vikram Iye	29	Male	West	Surat	Books	Self-Help E	2	949.5	0.22	2202.33	964.66	62.42	Cash on D	2	TRUE	Returned	8
ORD-0141	2021-09-28	CUST0245	Arjun Shal	31	Female	Central	Bhopal	Books	Self-Help E	3	329.03	0.02	994.87	375.07	77.34	Credit Car	5	TRUE	Returned	10
ORD-0291	2023-06-10	CUST2328	Aditya Sha	43	Female	North	Amritsar	Clothing	Saree	1	1356.6	0.01	2140.41	717.27	82.68	UPI	1	FALSE	Delivered	4
ORD-0046	2020-07-24	CUST2360	Aditya Pat	44	Female	East	Patna	Books	Self-Help E	8	495.77	0.37	2828.82	1338.88	84.5	Debit Card	1	FALSE	Cancelled	10
ORD-0010	2020-02-09	CUST7533	Arjun Das	17	FEMALE	East	Patna	Clothing	Kurta	9	2128.77	0.1	14568.2	4173.45	75.33	UPI	3	FALSE	Delivered	8
ORD-0036	2020-06-04	CUST8034	Meera Mis	51	Other	Central	Raipur	Furniture	Wardrobe	3	24460.8	0.15	86997.4	14762.2	60.11	UPI	1	FALSE	Delivered	8
ORD-0170	2022-01-28	CUST4777	Shreya Bo	14	Female	West	Ahmedaba	Groceries	Dal (1kg)	2	274.81	0.15	379.5	24.26	46.69	EMI	4	FALSE	Delivered	4
ORD-0227	2022-09-14	CUST3106	Vikram Pill	15	Other	North	Ludhiana	Beauty	Lipstick	7	111.14	0.37	567.33	305.65	44.94	Net Bankir	4	FALSE	Delivered	9
ORD-0341	2024-01-21	CUST8394	Karan Bos	29	Female	West	Pune	Books	Cook Book	10	1492.28	0.38	9315.19	4336.31	66.67	Debit Card	3	TRUE	Returned	
ORD-0253	2023-01-04	CUST7336	Deepa Josl	53	Male	South	Kochi	Clothing	T-Shirt	9	590.14		4824.04	2277.4	79.19	EMI	1	FALSE	Delivered	4
ORD-0352	2024-03-10	CUST5062	Deepa Me	30	Female	South	Kochi	Furniture	Chair	10	12138.9	0.15	144295	50995.3	49.93	EMI		FALSE	Delivered	1
ORD-0181	2022-03-14	CUST0879	Rahul Gup	26	Male	South	Bangalore	Clothing	Dress	10	1706.05	0.04	17995.4	6453.4	89.15	UPI	5	FALSE	Delivered	4
ORD-0325	06/11/2023	CUST8825	Shreya Sin	15	Male	East	Guwahati	Electronics	Tablet	8	73558.4	0.4	315130	53808.7	82.71	UPI	3	FALSE	Delivered	6

Data Cleaning and Transformation

Duplicate row identified: ORD-00881 appeared 2 times in the dataset and one occurrence was removed.

Select all data → Go to the Data tab → Click Remove Duplicates, limiting the dataset to 600 unique rows including headers.



Date Format: For Order Date column, the dataset has mixed date formats such as:

- 2024-08-15
- 02/09/2024
- 17-Aug-24
- 08-Apr-20

Order date values were standardized by converting multiple date formats into a consistent and uniform date format (DD-MM-YYYY). Standardized the Order Date column by using Excel's Text to Columns feature: Select Date Column → Text to Columns → Delimited → No Delimiters → Date (DMY) → Finish. This converted all mixed date values into a consistent DD-MM-YYYY format.

order_date	DD-YY-MMM
2024-08-15	15-08-2024
2022-08-16	16-08-2022
2021-08-29	29-08-2021
2022-07-05	05-07-2022
2020-05-07	07-05-2020
02/09/2024	02-09-2024
2021-09-28	28-09-2021

Handling Inconsistencies and Missing Values:

Age column imputation: Identified invalid values in the Age column, such as negative numbers (-6, -5) and unrealistic values (999). These values were considered invalid, replaced with blank cells, and handled during the missing value treatment process to improve data accuracy and consistency.

Used the following Excel formula to replace blank values in the Age column with the average age of the dataset: Syntax:

=IF(ISBLANK(F2),AVERAGE(F:F),F2)

C	D	E	F	G
Y-MI	custom	custom	age	age imputation
8-2024	CUST3260	Aarav Mel	39	39
8-2022	CUST0788	Sanjay Ver	28	28
8-2021	CUST4929	Pooja Shai	41	41
7-2022	CUST0558	Priya Redc	29	29
5-2020	CUST5803	Ritu Sharn	9	9

Age Group column: A new calculated column, Age Group, was created using IF function to classify customers into the following age ranges:

- Child: Less than 13 years
- Teen: 13–19 years
- Young Adult: 20–35 years
- Adult: 36–50 years
- Senior: 51–65 years
- Elderly: Above 65 years

The Age Group column was created to group customers into age ranges for easier analysis of sales and order patterns by customer segment.

Syntax:

```
=IF(G2<13,"Child",IF(G2<=19,"Teen",IF(G2<=35,"Young Adult",IF(G2<=50,"Adult",IF(G2<=65,"Senior",IF(G2>66,"Elderly"))))))
```

age imp	age group
39	Adult
28	Young Adult
41	Adult
29	Young Adult
9	Child

Gender Column: Standardized format using Proper Function

Standardized the gender values by replacing M, male, and MALE with Male, and F, f, and female with Female.
Syntax: =PROPER(I2)

MALE	Male
MALE	Male
Other	Other
Female	Female
Other	Other
Other	Other

Quantity column Imputation: Invalid values (-1 and 0) were replaced with blank values, and the blank values were filled using the average quantity. Average value is 7.

Syntax: =IF(ISBLANK(O2), AVERAGE(O:O), O2)

quantit	quantit
	7
	7
	7
	7
	7
	7

Discount_Pct column: Blank values were replaced with the average discount percentage.

Syntax: =IF(ISBLANK(R2), AVERAGE(R:R),R2)

discour	discour
	20%
0.08	8%
0.14	14%
0.17	17%
0.31	31%
0.22	22%

Customer Satisfaction column: Blank values were identified and replaced with the average customer satisfaction rating. Average value is 3.

Syntax: =IF(ISBLANK(Y2), AVERAGE(Y:Y),Y2)

5	5
2	2
	3
	3
2	2
5	5

Order Status Column: Standardized format using Proper Function by changing delivered and Delivered to Delivered, and cancelled and Cancelled to Cancelled, ensuring consistent capitalization.

Syntax: =PROPER(AB2)

Cancelled	Cancelled
cancelled	Cancelled
Cancelled	Cancelled
cancelled	Cancelled
Cancelled	Cancelled

Days to Ship column: Invalid values such as -1, 0, and 45(outlier) were replaced with blank values, and the blank values were then replaced with the average Days to Ship value. The average value is 5.

Syntax: =IF(ISBLANK(AD10),MEDIAN(AD:AD),AD10)

8	8
4	4
9	9
	5
4	4
1	1
4	4

Profit Margin calculated column: Created by dividing Profit by Sales Amount and formatting the result as a percentage.

Syntax: =Profit/Sales Amount

The Profit Margin column was created to measure the percentage of sales amount retained as profit. It helps compare profitability across different products, categories, or orders.

profit mrgin
7.15%
58.55%
33.55%
25.43%
9.69%
43.80%

Cell Formatting: Format Cells was applied to the relevant columns to ensure consistent formatting and data types. Unit Price, Sales Amount, Profit and Shipping Cost columns were formatted as currency. Discount pct column Values were formatted as percentages to ensure consistent percentage representation.

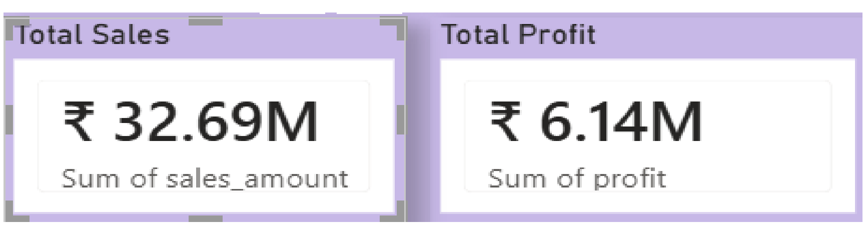
unit_price	discount_	discount_pct_ in	sales_amount	profit	profit mrgin	shipping_cost
₹ 346.32		20%	₹ 4,043.14	₹ 289.18	7.15%	₹ 56.04
₹ 858.66	0.08	8%	₹ 9,147.31	₹ 5,355.56	58.55%	₹ 64.49
₹ 25,941.32	0.14	14%	₹ 110,156.97	₹ 36,956.51	33.55%	₹ 67.65
₹ 22,071.92	0.17	17%	₹ 262,066.11	₹ 66,644.99	25.43%	₹ 54.47
₹ 320.06	0.31	31%	₹ 323.95	₹ 31.38	9.69%	₹ 48.81
₹ 949.50	0.22	22%	₹ 2,202.33	₹ 964.66	43.80%	₹ 62.42

Cleaned Dataset: Using XLOOKUP, a cleaned data sheet was created by retrieving the required values from the original dataset and consolidating them into a structured table using Order ID as the unique identifier.

Order Date	Customer	Customer	Age	Age Group	Gender	Region	City	Product C	Product N	Quantity	Unit Price	Discount_	Sales Amount	Profit	Profit Marg	Shipping Cost	Payment M	Customer	Return Fla	Order Stat	Days to Ship
03-01-2020	CUST6495	Nisha Meh	23	Young Adu	Female	South	Hyderabad	Furniture	Sofa	8	₹ 22,523.15	34%	₹ 95,036.79	₹ 29,818.97	31.38%	₹ 65.72	EMI	2	FALSE	Delivered	5
08-01-2020	CUST0587	Nisha Nair	12	Child	Other	Central	Nagpur	Sports	Dumbbells	2	₹ 2,448.88	8%	₹ 3,192.22	₹ 355.63	11.14%	₹ 47.54	Cash on D	3	FALSE	Delivered	9
10-01-2020	CUST9137	Karan Iyer	52	Senior	Female	West	Mumbai	Books	Cook Bool	10	₹ 324.41	32%	₹ 1,810.18	₹ 903.99	49.94%	₹ 48.24	UPI	3	FALSE	Delivered	4
11-01-2020	CUST0340	Karan Misl	23	Young Adu	Male	Central	Indore	Sports	Football	7	₹ 12,178.99	1%	₹ 66,009.55	₹ 17,076.44	25.87%	₹ 51.51	UPI	5	FALSE	Cancelled	10
12-01-2020	CUST2161	Arjun Das	49	Adult	Other	West	Mumbai	Groceries	Flour	7	₹ 315.85	13%	₹ 203.41	₹ 13.82	6.79%	₹ 60.46	Cash on D	5	FALSE	Delivered	10
14-01-2020	CUST7922	Sunita Rec	20	Young Adu	Male	East	Kolkata	Books	Fiction No	1	₹ 1,172.06	18%	₹ 613.66	₹ 243.99	39.76%	₹ 62.05	Cash on D	4	FALSE	Delivered	1
19-01-2020	CUST4340	Sanjay Bos	21	Young Adu	Other	North	Delhi	Furniture	Sofa	3	₹ 34,890.63	29%	₹ 64,727.53	₹ 18,067.63	27.91%	₹ 40.97	Net Bankii	5	FALSE	Shipped	3
20-01-2020	CUST2797	Shreya Pill	13	Teen	Male	South	Hyderabad	Beauty	Perfume	6	₹ 1,952.59	14%	₹ 6,658.21	₹ 3,867.94	58.09%	₹ 38.17	Credit Car	1	FALSE	Shipped	8
21-01-2020	CUST0719	Deepa Kap	33	Young Adu	Other	East	Guwahati	Beauty	Lipstick	9	₹ 1,931.61	35%	₹ 10,532.18	₹ 5,257.88	49.92%	₹ 98.73	Net Bankii	5	FALSE	Delivered	3
22-01-2020	CUST6012	Ritu Kapoc	16	Teen	Other	South	Bangalore	Books	Textbook	4	₹ 363.32	7%	₹ 1,026.50	₹ 410.28	39.97%	₹ 89.93	EMI	5	FALSE	Shipped	5
28-01-2020	CUST4466	Ananya Na	34	Young Adu	Male	North	Delhi	Sports	Dumbbells	4	₹ 12,945.28	13%	₹ 31,129.56	₹ 7,042.29	22.62%	₹ 65.38	Credit Car	5	FALSE	Delivered	4
03-02-2020	CUST6388	Amit Das	29	Young Adu	Female	West	Ahmedaba	Furniture	Chair	10	₹ 9,468.07	32%	₹ 57,174.24	₹ 15,202.22	26.59%	₹ 40.98	EMI	3	FALSE	Delivered	2
06-02-2020	CUST2874	Kavya Bha	38	Adult	Female	East	Kolkata	Beauty	Hair Oil	3	₹ 1,535.87	4%	₹ 3,186.03	₹ 1,346.80	42.27%	₹ 86.51	Cash on D	5	TRUE	Returned	7

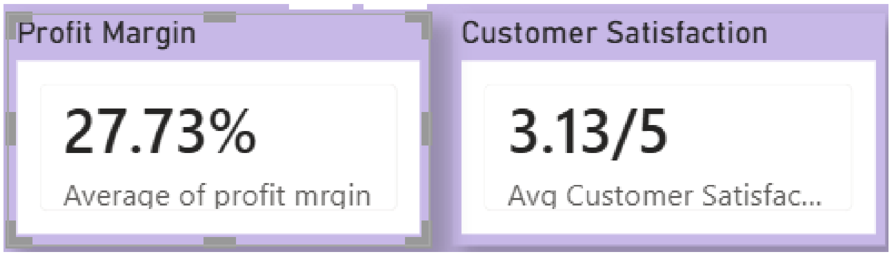
Data Visualization

Cleaned data was imported into Power BI for data analysis and dashboard creation.



• Total Sales Card: A card visual was created to display the total sales amount by using the Sum of Sales Amount aggregation.

• Total Profit Card: A separate card visual was created to display the total profit by using the Sum of Profit aggregation.

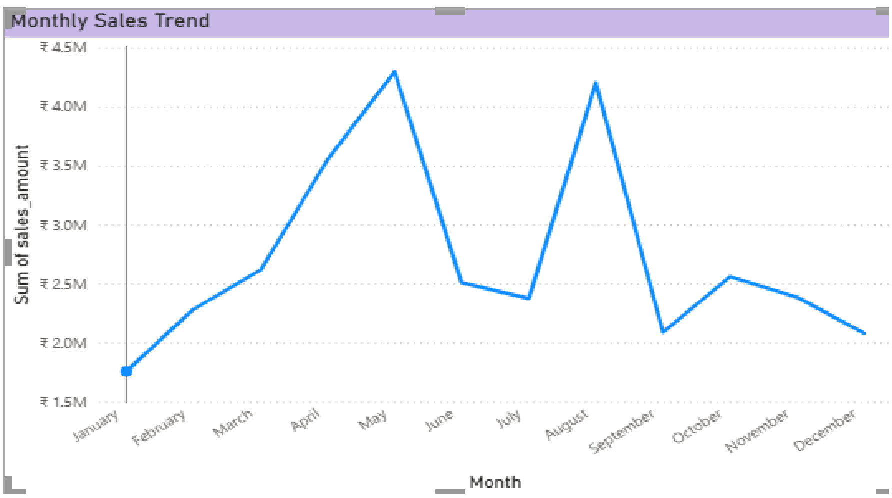


- Average Profit Margin Card: A card visual was created to display the average profit margin across all orders.

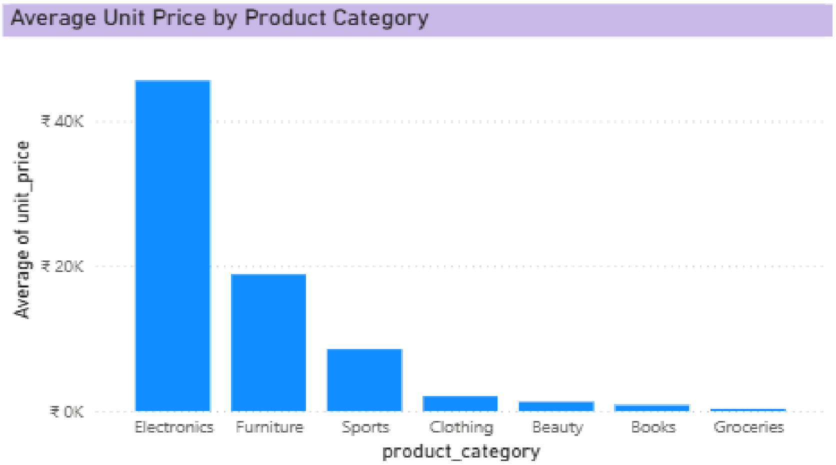
- Average Customer Satisfaction Card: A DAX measure was created to calculate the average customer satisfaction rating:

```
Avg Customer Satisfaction = FORMAT(AVERAGE('Retail Sales'[customer_satisfaction]),"0.00")&"/5"
```

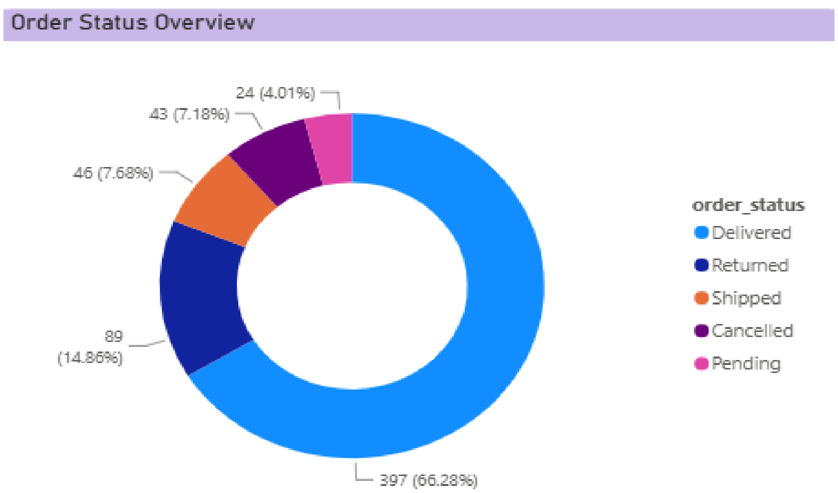
The measure was then added to a card visual to display the average customer satisfaction rating.



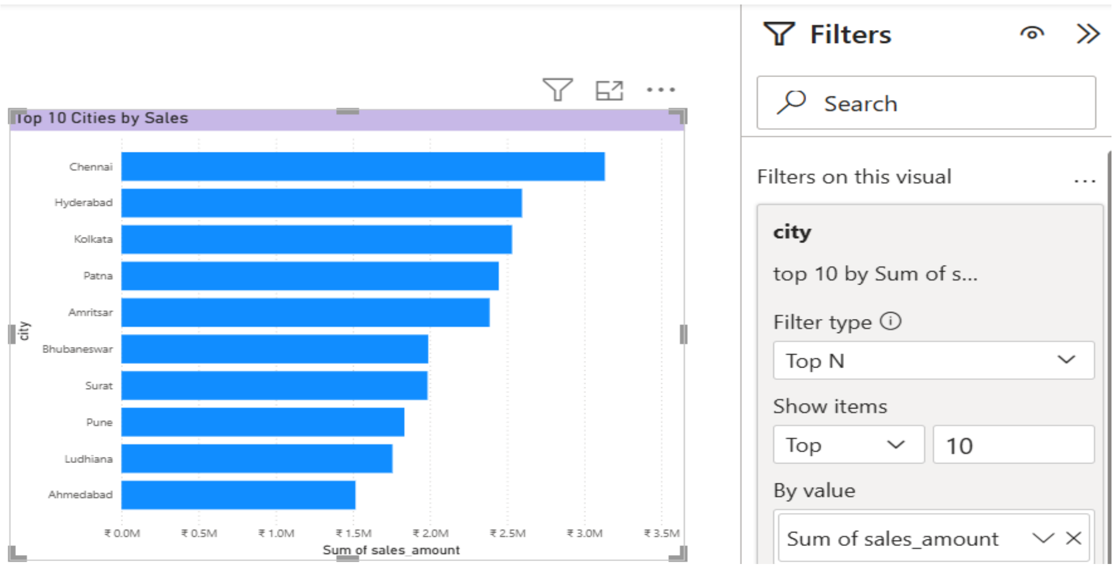
Monthly Sales Trend: A line chart was created to visualize the trend of total sales across different months. Order Date was used on the X-axis, and Sum of Sales Amount was used on the Y-axis.



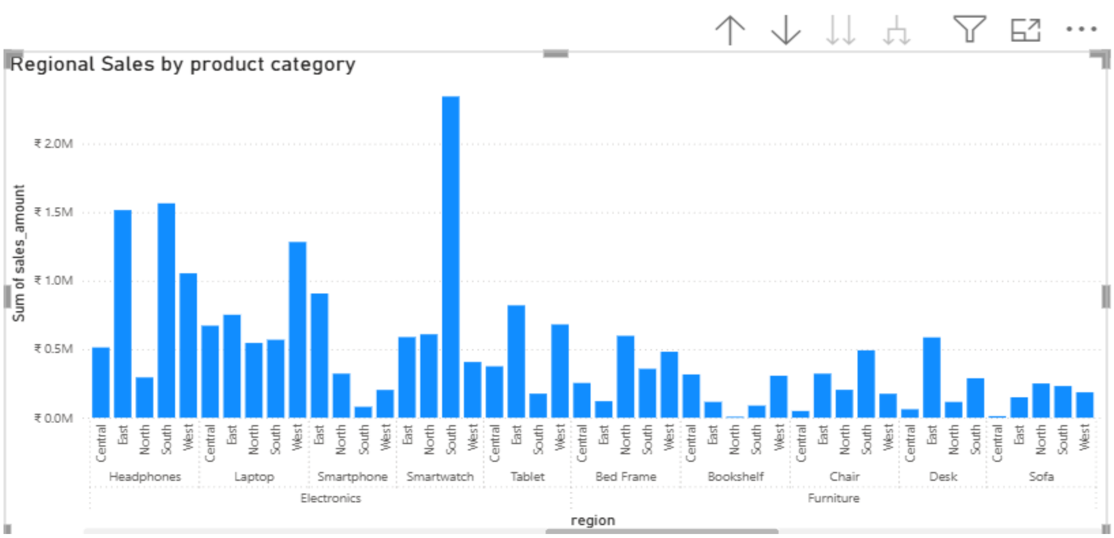
Average Unit Price by Product Category: A column chart was created to compare the average unit price across different product categories. Product Category was used on the X-axis, and Average of Unit Price was used on the Y-axis. It helps identify which product categories have higher or lower average unit prices and supports pricing and product analysis.



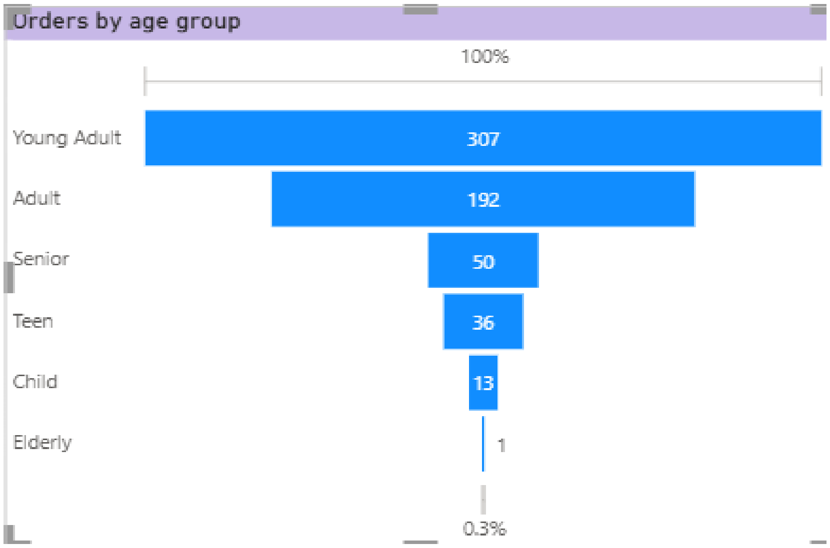
Order Status Overview: A donut chart was created to show the distribution of orders across different order statuses. It helps identify how many orders are delivered, returned, shipped, or cancelled, providing an overview of the overall order performance.



Top 10 Cities by Sales: A bar chart was created to display the top 10 cities based on total sales amount. A Top N filter was applied to the City field, with N set to 10 and Total Sales used to rank the cities. This helps identify the cities generating the highest sales.

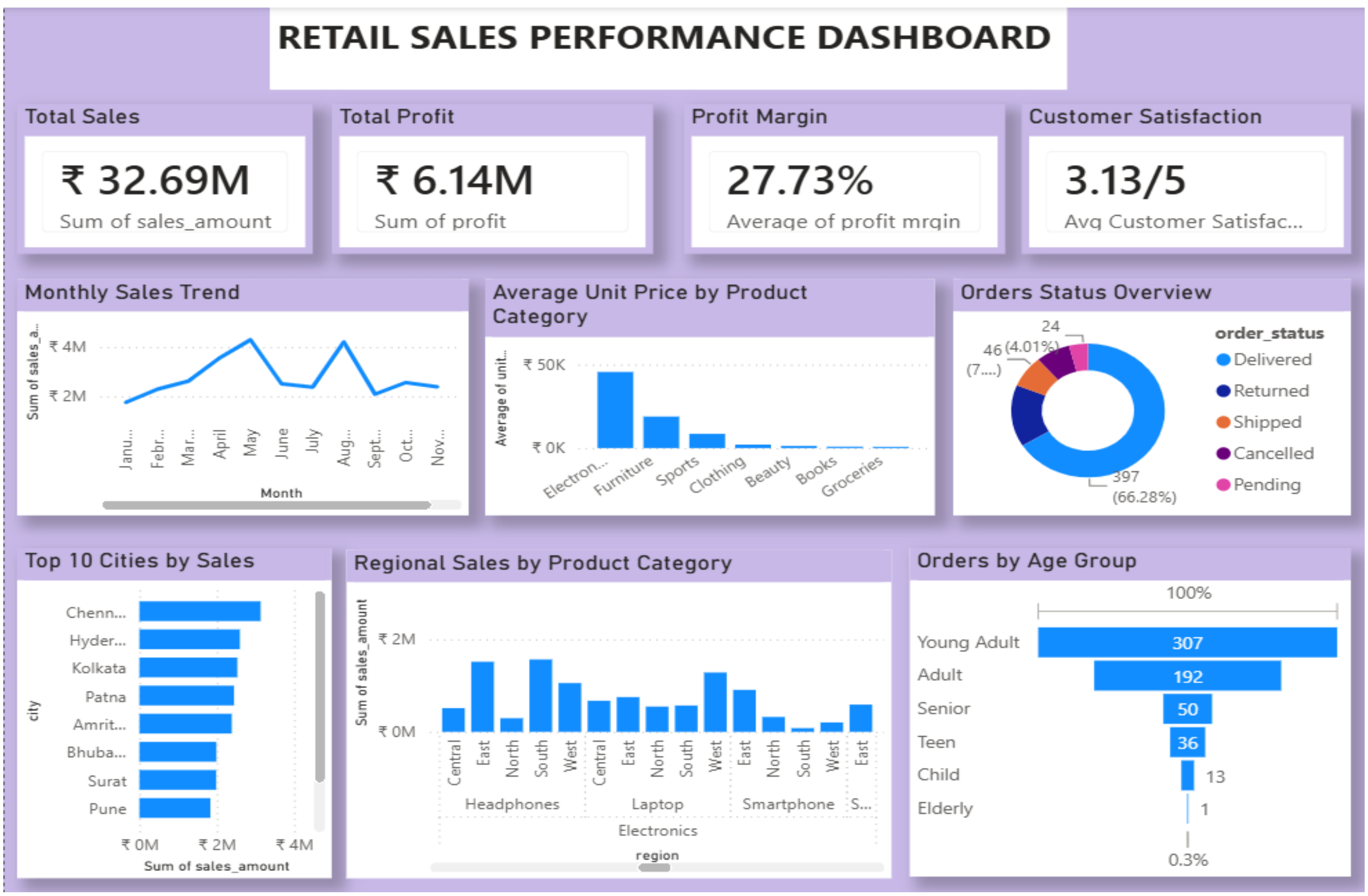


Regional Sales by Product Category: A hierarchy was created using Product Category, Product Name, and Region. This hierarchy was used to analyze sales across product categories, individual products, and regions, allowing detailed drill-down analysis of sales performance using clustered column chart.



Orders by Age Group: A funnel chart was created to show the distribution of orders across different age groups. The Age Group field was used as the category, and Count of Order ID was used as the value. This helps compare the number of orders placed by each age group.

Dashboard:



Dashboard Overview:

The dashboard provides a comprehensive overview of sales performance, profitability, customer satisfaction, and order trends. Key performance indicators such as Total Sales, Total Profit, Average Profit Margin, and Average Customer Satisfaction provide a quick view of overall business performance. The dashboard also analyzes monthly sales trends, order status distribution, top 10 cities by sales, average unit price across product categories, and regional sales performance. In addition, the age-group analysis provides insights into customer ordering patterns. Overall, the dashboard helps identify key sales trends, high-performing regions and cities, product pricing patterns, and customer behavior to support data-driven business decisions.

Summary

The project involved cleaning, transforming, analyzing, and visualizing retail sales data using Microsoft Excel and Power BI. In Excel, the dataset was cleaned by removing duplicates, identifying and correcting invalid values, handling blank and missing values, standardizing date and text formats, and applying appropriate currency and percentage formats. Calculated columns such as Profit Margin were created, and XLOOKUP was used with Order ID as the unique identifier to prepare a structured cleaned data sheet.

The cleaned data was then imported into Power BI for analysis and dashboard development. Key performance indicators including Total Sales, Total Profit, Average Profit Margin, and Average Customer Satisfaction were displayed using card visuals. DAX was used to create the Average Customer Satisfaction measure. Various visualizations were created to analyze monthly sales trends, order status distribution, average unit price by product category, top 10 cities by sales, regional sales by product category, and orders by age group.

Overall, the project demonstrates the complete data analysis process, from data cleaning and transformation in Excel to interactive visualization and business analysis in Power BI. The dashboard provides meaningful insights into sales performance, profitability, customer satisfaction, product pricing, regional performance, order fulfillment, and customer behavior, supporting data-driven decision-making.